

Lecture 1: Electrical Quantities and Applications

Basic SI Units; Charge, Force, Work and Power;
Electrical Potential, EMF and Energy

EET101: ELECTRONICS

Course:	Pre-requisites:
EET101 – Electronics	None (First Year, Semester 1)
Teaching Hours:	Assessment:
45 contact hours per semester (~3 hrs/week)	70% Final Exam, 30% CATs & Assignments

Abstract

This first lecture lays the quantitative foundation for the whole of EET101. It begins by placing electronics within the International System of Units (SI), establishing the base and derived units that every later calculation in the course depends on. The lecture then defines electric charge as the fundamental electrical quantity, links it to current through the relation $Q = It$, and introduces Coulomb's law as the force of interaction between charges. Work and energy are then developed from first principles of mechanics and re-expressed in electrical terms, leading naturally to the definitions of electric potential, potential difference and electromotive force (emf). The lecture closes by defining electrical power and energy and relating them to the practical unit of electricity billing, the kilowatt-hour. Worked examples throughout show how these quantities are calculated and combined, preparing the ground for Ohm's law, network theorems and component behaviour in the lectures that follow.

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1 Learning Objectives

Upon completion of this lecture, students should be able to:

1. State the seven SI base quantities and units, and identify the derived units used throughout electronics (coulomb, volt, joule, watt).
2. Define electric charge, state the value of the elementary charge, and relate charge to current through $Q = It$.
3. State and apply Coulomb's law to calculate the electrostatic force between two point charges.
4. Define work and energy, and derive the electrical expression for work done in moving a charge through a potential difference, $W = QV$.
5. Define electric potential, potential difference and electromotive force (emf), and distinguish emf from terminal (potential difference) voltage.
6. Define electrical power, relate it to voltage, current and resistance ($P = VI = I^2R = V^2/R$), and calculate energy consumed as $W = Pt$.
7. Convert between joules and kilowatt-hours, and perform basic cost-of-electricity calculations.
8. Solve numerical problems combining charge, force, work, potential, emf, power and energy.

How This Lecture Is Organised

The lecture develops the material in three stages:

1. **Part I — Units and Measurement.** Why electrical quantities need to be defined precisely, and the SI system that underlies every calculation in the course (Sections 2–3).
2. **Part II — Charge and Force.** Electric charge as the fundamental electrical quantity, its quantisation, its relation to current, and the force it exerts on other charges (Sections 4–5).
3. **Part III — Work, Potential, EMF, Energy and Power.** How moving charge against a field requires work, how this defines potential and emf, and how power and energy follow (Sections 6–9).

Part I — Units and Measurement

2 Introduction: Why Electrical Quantities Matter

Every branch of electrical and electronic engineering – from the design of a battery-powered sensor to the operation of a national power grid – rests on being able to measure and calculate a small set of physical quantities precisely: charge, current, force, work, energy, potential and power. Before any circuit law (Ohm's law, Kirchhoff's laws) can be applied meaningfully, the quantities and units in which it is expressed must be agreed and consistent. This lecture establishes that common language.

Engineering calculations are only as trustworthy as the units behind them. A numerical answer without a correctly stated unit – or with units mixed inconsistently (for example, mixing

millivolts and kilo-ohms without converting) – is not a complete engineering answer and is a common source of error in both examinations and real designs.

Fundamental Insight: Why Start With Units?

Physical laws such as $F = QE$ or $P = VI$ are only true when every quantity is expressed in a consistent system of units. Engineers worldwide use the International System of Units (SI) for exactly this reason: it guarantees that a calculation performed in Nairobi and one performed in Tokyo, using the same SI values, produce the same numerical result.

3 The International System of Units (SI)

3.1 Base Quantities and Units

The SI system defines seven base quantities, each with its own base unit. All other physical quantities used in electronics – including charge, potential, energy and power – are derived from combinations of these base units.

Base Quantity	SI Unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

Table 1: The seven SI base quantities and units

3.2 Derived Electrical Units

Of the seven base units, the ampere (A) is the only one defined directly in electrical terms; every other electrical unit used in this course is derived from it together with the metre, kilogram and second.

Quantity	Unit	Symbol	In Base/Derived Units
Charge	coulomb	C	$1 \text{ C} = 1 \text{ A} \cdot \text{s}$
Force	newton	N	$1 \text{ N} = 1 \text{ kg} \cdot \text{m} \cdot \text{s}^{-2}$
Work / Energy	joule	J	$1 \text{ J} = 1 \text{ N} \cdot \text{m} = 1 \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-2}$
Power	watt	W	$1 \text{ W} = 1 \text{ J} \cdot \text{s}^{-1}$
Potential difference / EMF	volt	V	$1 \text{ V} = 1 \text{ J} \cdot \text{C}^{-1}$
Resistance	ohm	Ω	$1 \Omega = 1 \text{ V} \cdot \text{A}^{-1}$

Table 2: Derived electrical units used throughout EET101, expressed in base SI units

Common Pitfall

The volt is *not* a base unit – it is defined as one joule per coulomb ($1 \text{ V} = 1 \text{ J/C}$). Students sometimes treat the volt as though it were a fundamental, independent quantity; in fact it is entirely defined by energy and charge, which is exactly why the relation $W = QV$ (Section 7) is not an approximation but a restatement of what a volt means.

4 Electric Charge

Electric charge, symbol Q (or q), is the fundamental property of matter that gives rise to all electrical phenomena. Charge occurs in two kinds, conventionally labelled positive and negative; like charges repel and unlike charges attract. The SI unit of charge is the **coulomb (C)**, defined as the amount of charge transferred by a steady current of one ampere flowing for one second.

4.1 Quantisation of Charge and the Elementary Charge

Charge is **quantised**: it exists only in integer multiples of the elementary charge, e , carried by a single electron (negative) or proton (positive).

$$e = 1.602 \times 10^{-19} \text{ C.}$$

The charge on any object is therefore $Q = ne$, where n is an integer number of electrons transferred (n is enormous for everyday charges – a charge of just 1 C corresponds to about 6.24×10^{18} electrons – which is why charge appears continuous at the macroscopic scale used in circuit calculations).

4.2 Current as the Rate of Flow of Charge

Electric current, I , is defined as the rate of flow of charge past a point in a circuit:

$$I = \frac{Q}{t} \quad (\text{for a steady current}), \quad i = \frac{dq}{dt} \quad (\text{instantaneously}).$$

Rearranged, this gives the amount of charge transferred by a steady current over a time t :

$$\boxed{Q = It}$$

This single relation is the bridge between the two quantities used throughout electronics: the current we measure with an ammeter, and the charge that current has transported after a given time – the quantity that, for example, determines how long a battery of known ampere-hour rating will last under a given load.

5 Electric Force — Coulomb's Law

Two point charges exert a force on one another along the line joining them. This force is described by **Coulomb's law**, one of the foundational results of electrostatics: the force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them.

$$\boxed{F = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{d^2}}$$

where Q_1 and Q_2 are the two charges (in coulombs), d is the separation between them (in metres), and ϵ_0 is the permittivity of free space, $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$. The constant $1/4\pi\epsilon_0$ is often written k and has the approximate value $k \approx 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$. The force is repulsive if the two charges have the same sign, and attractive if they have opposite signs, and it acts along the straight line joining the two charges, as shown in Figure 1.

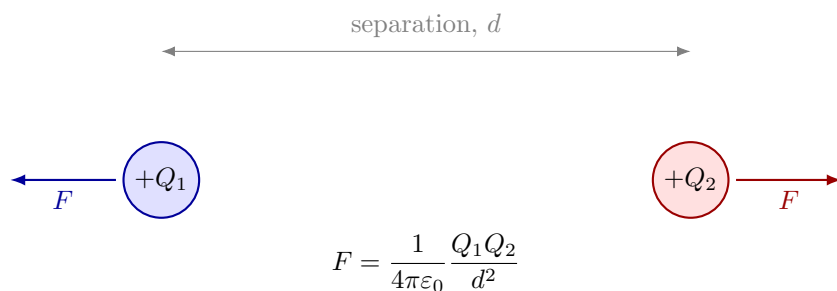


Figure 1: Coulomb’s law: two like charges Q_1 and Q_2 separated by distance d experience an equal and opposite repulsive force F along the line joining them

Fundamental Insight: Coulomb’s Law and Newton’s Third Law

The two forces shown in Figure 1 are equal in magnitude and opposite in direction – this is simply Newton’s third law applied to an electrical interaction. Coulomb’s law describes the force between charges at rest (electrostatics); once charges are in motion, as in a current-carrying conductor, additional magnetic forces also arise, which are outside the scope of this introductory lecture.

Part III — Work, Potential, EMF, Energy and Power

6 Work and Energy in Moving a Charge

From mechanics, work is done whenever a force moves its point of application through a distance: $W = Fd$ (for a constant force acting along the direction of motion). Energy is the capacity to do work, and both work and energy share the SI unit of the joule (J).

In an electric field, moving a charge Q against the field requires work to be done against the electrical force acting on it. This work does not disappear – it is stored as electrical potential energy, which can later be recovered (for example, as the charge is allowed to move back under the field, or as current is driven around a circuit).

7 Electric Potential, Potential Difference and EMF

The work done per unit charge in moving a charge between two points defines the **potential difference** between those points:

$$V = \frac{W}{Q} \quad \Longleftrightarrow \quad \boxed{W = QV}$$

The SI unit of potential difference is the **volt (V)**, where 1 volt is defined as a potential difference of 1 joule per coulomb. Potential difference (p.d.), often loosely called “voltage”, is what drives current around a circuit: current flows from a point of higher potential to a point of lower potential through an external circuit.

Electromotive force (emf), symbol ε (or E), is the energy supplied per unit charge by a source – a cell, battery, generator or solar panel – in driving charge around a complete circuit. Like potential difference, emf is measured in volts. However, emf and terminal potential

difference are *not* the same quantity once current flows, because every practical source has some internal resistance, r , across which some of the emf is “used up” internally:

$$V = \varepsilon - Ir$$

where V is the terminal (potential-difference) voltage available to the external circuit, ε is the emf, I is the current drawn, and r is the source’s internal resistance. Only when no current is drawn (open circuit, $I = 0$) does the terminal voltage equal the full emf.

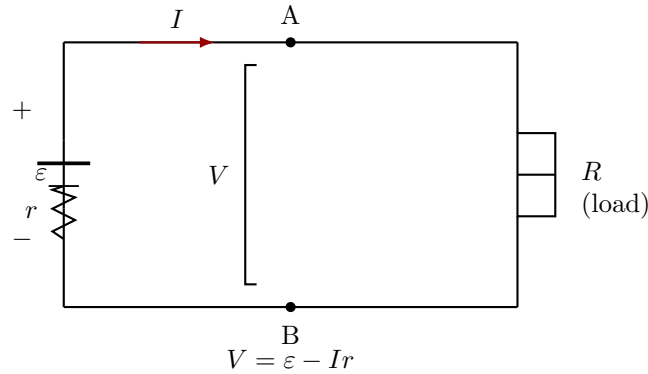


Figure 2: A source of emf ε with internal resistance r , supplying current I to a load R . The terminal voltage V measured between A and B is less than the emf whenever current flows

Common Pitfall

Emf and terminal voltage are frequently confused. The emf, ε , is a fixed property of the source (how much energy it supplies per coulomb of charge it drives around the circuit). The terminal voltage, V , is what a voltmeter actually reads across the source’s terminals, and it falls below ε as soon as current is drawn, because of the voltage dropped across the source’s own internal resistance, r . Always check whether a problem is asking for ε or for V .

8 Electrical Power and Energy

Power is the rate at which work is done, or equivalently the rate at which energy is transferred or converted:

$$P = \frac{W}{t}.$$

For an electrical circuit, substituting $W = QV$ and $Q = It$ gives the standard electrical power relations, all of which are equivalent and interchangeable using $V = IR$ (Ohm’s law, formally introduced in Lecture 4):

$$P = VI = I^2R = \frac{V^2}{R}$$

The SI unit of power is the **watt (W)**, where 1 watt = 1 joule per second. Rearranging $P = W/t$ gives the energy consumed (or supplied) over a time t :

$$W = Pt$$

When t is in seconds and P is in watts, W comes out in joules. In practice, electrical energy is billed in a much larger, more convenient unit – the **kilowatt-hour (kWh)**, sometimes called a “unit” of electricity – defined as the energy consumed by a load of 1 kW operating for 1 hour:

$$1 \text{ kWh} = 1000 \text{ W} \times 3600 \text{ s} = 3.6 \times 10^6 \text{ J}.$$

Fundamental Insight: Why Two Different Energy Units?

The joule is the correct SI unit for energy and is essential for consistent calculation. The kilowatt-hour exists purely for convenience: household and industrial energy consumption over hours or days, expressed in joules, would involve inconveniently large numbers (a single 1 kW kettle used for 10 minutes already consumes 6×10^5 J). Utility companies therefore bill in kWh, and an engineer must be able to convert confidently between the two.

9 Worked Examples

Worked Example

Example 1: Charge, Current and Number of Electrons

A steady current of 2.5 A flows through a conductor for 4 minutes. Determine (a) the charge transferred, and (b) the number of electrons that pass a given cross-section of the conductor in this time.

Solution

(a) Using $Q = It$, with t converted to seconds ($t = 4 \times 60 = 240$ s):

$$Q = 2.5 \times 240 = 600 \text{ C.}$$

(b) Each electron carries charge $e = 1.602 \times 10^{-19}$ C, so the number of electrons is:

$$n = \frac{Q}{e} = \frac{600}{1.602 \times 10^{-19}} = 3.75 \times 10^{21} \text{ electrons.}$$

Engineering interpretation: even a modest current sustained for a few minutes involves an astronomically large number of individual electrons, which is why charge is treated as a continuous quantity in ordinary circuit analysis.

Worked Example

Example 2: Coulomb's Law — Force Between Two Charges

Two small charged spheres carry charges of $+3 \mu\text{C}$ and $+5 \mu\text{C}$ respectively and are placed 20 cm apart in air. Calculate the force between them and state its nature.

Solution

Using Coulomb's law, $F = (1/4\pi\epsilon_0)(Q_1Q_2/d^2)$, with $k = 1/4\pi\epsilon_0 \approx 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$:

$$F = 8.99 \times 10^9 \times \frac{(3 \times 10^{-6})(5 \times 10^{-6})}{(0.20)^2}$$

$$F = 8.99 \times 10^9 \times \frac{1.5 \times 10^{-11}}{0.04} = 3.37 \text{ N.}$$

Since both charges are positive (like charges), the 3.37 N force is **repulsive** – each sphere pushes the other away along the line joining them.

Worked Example

Example 3: Work Done Moving a Charge Through a Potential Difference

A charge of 40 mC is moved between two points in a circuit across which the potential difference is 12 V. Calculate the work done, and state where this energy is transferred to.

Solution

Using $W = QV$, with $Q = 40 \times 10^{-3} \text{ C}$ and $V = 12 \text{ V}$:

$$W = 40 \times 10^{-3} \times 12 = 0.48 \text{ J.}$$

This 0.48 J of energy, supplied by whatever drives the charge between the two points, is transferred to the circuit element between them – for example, dissipated as heat in a resistor, or stored as chemical or magnetic energy in other components (Lectures 4 and 7).

Worked Example**Example 4: EMF, Internal Resistance and Terminal Voltage**

A battery has an emf of 12.0 V and an internal resistance of 0.50 Ω . It supplies a current of 3.0 A to an external load. Calculate (a) the terminal voltage of the battery, and (b) the power dissipated inside the battery itself due to its internal resistance.

Solution

(a) Using $V = \varepsilon - Ir$:

$$V = 12.0 - (3.0 \times 0.50) = 12.0 - 1.5 = 10.5 \text{ V.}$$

(b) The power dissipated internally is found from $P = I^2r$, using the internal resistance:

$$P = (3.0)^2 \times 0.50 = 9 \times 0.50 = 4.5 \text{ W.}$$

Engineering interpretation: 1.5 V of the battery's 12.0 V emf is “lost” internally, and 4.5 W is wasted as heat inside the battery rather than delivered to the load – this is precisely why a battery with low internal resistance is preferred for high-current applications.

Worked Example**Example 5: Electrical Power, Energy and Cost**

An electric heater is rated at 2 kW and is operated from a 240 V supply for 3 hours each day. Determine (a) the current drawn by the heater, (b) the energy consumed per day in kWh, and (c) the cost of running the heater for 30 days if electricity costs KES 25 per kWh.

Solution

(a) Using $P = VI$, rearranged for I :

$$I = \frac{P}{V} = \frac{2000}{240} = 8.33 \text{ A.}$$

(b) Using $W = Pt$, with $P = 2 \text{ kW}$ and $t = 3 \text{ h}$:

$$W = 2 \times 3 = 6 \text{ kWh per day.}$$

(c) Over 30 days, the total energy consumed and its cost are:

$$\text{Total energy} = 6 \times 30 = 180 \text{ kWh,}$$

$$\text{Cost} = 180 \times 25 = \text{KES } 4500.$$

Engineering interpretation: everyday energy billing is simply an application of $W = Pt$, expressed in kWh rather than joules for convenience (Section 8).

10 Lecture Summary

The key ideas from Lecture 1 are:

1. All electrical calculations rest on the SI system: the ampere is the only base electrical unit, and charge (C), potential/emf (V), work/energy (J) and power (W) are all derived from it.
2. Electric charge is quantised in units of the elementary charge, $e = 1.602 \times 10^{-19}$ C, and relates to current through $Q = It$.
3. Coulomb's law, $F = (1/4\pi\epsilon_0)(Q_1Q_2/d^2)$, gives the force between two point charges: repulsive for like charges, attractive for unlike charges.
4. Work done moving a charge Q through a potential difference V is $W = QV$; this defines the volt as one joule per coulomb.
5. Electromotive force (emf, ε) is the energy supplied per unit charge by a source; terminal voltage $V = \varepsilon - Ir$ is always less than ε once current I flows through a source of internal resistance r .
6. Electrical power is $P = VI = I^2R = V^2/R$, and energy consumed is $W = Pt$; the practical billing unit of electricity is the kilowatt-hour, $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$.

Lecture 2 continues with cells and batteries (electrolysis, construction, primary and secondary cells) and passive components (resistors, capacitors and inductors), building directly on the charge, potential and energy concepts established here.

11 Recommended Reading

1. Nilsson, J. W., & Riedel, S. (2009), *Electric Circuits* (8th Edition), Prentice Hall.
2. Hayt, W. H., Kemmerly, J., & Durbin, S. M. (2009), *Engineering Circuit Analysis* (7th Edition), McGraw-Hill.
3. Floyd, T. L. (2010), *Electronics Fundamentals: Circuits, Devices & Applications* (8th Edition).
4. Mehta, V. K., & Mehta, R. (1993), *Principles of Electronics*, S. Chand.
5. Svoboda, J. A., & Dorf, R. C. (2007), *Introduction to Electric Circuits*, John Wiley & Sons Inc.